Name _____ Period _____ Date _____

3-1 Lesson Quiz: Graphing Polynomial Functions

short form

Question numbers are the Savvas quiz numbers; questions 1 and 5 are not on this form.

2. Use the leading coefficient and degree of the polynomial function $f(x) = x^3 - 7x^2 + 10x$ to determine the end behavior of its graph. Select **all** the true statements.

☐ **A.** As $x \to \infty$, $y \to \infty$.

☐ **B.** As $x \to \infty$, $y \to -\infty$.

☐ **C.** As $x \to \infty$, $y \to 0$.

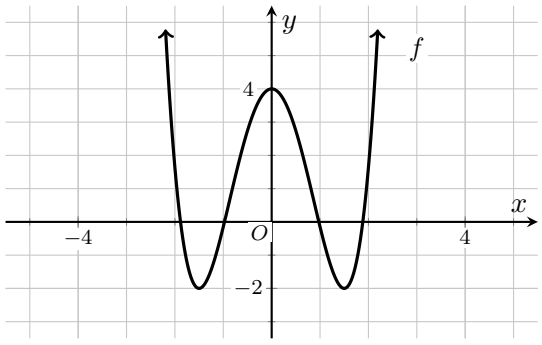
☐ **D.** As $x \to -\infty$, $y \to \infty$.

☐ **E.** As $x \to -\infty$, $y \to -\infty$.

Degree: _____ Leading coefficient: _____

3. The graph of a function f is shown. Use the graph to estimate all turning points. Select **all** x-values that are turning points of f.

☐ **A.** -3
☐ **B.** 0
☐ **C.** 4
☐ **D.** -1.5
☐ **E.** 1
☐ **F.** -1
☐ **G.** 1.5



4. Use the graph of the polynomial function $f(x) = -x^3 + 5x^2 - 2x - 8$ to complete the sentences. Its turning points are at $x = a \approx 0.21$ and $x = b \approx 3.12$.

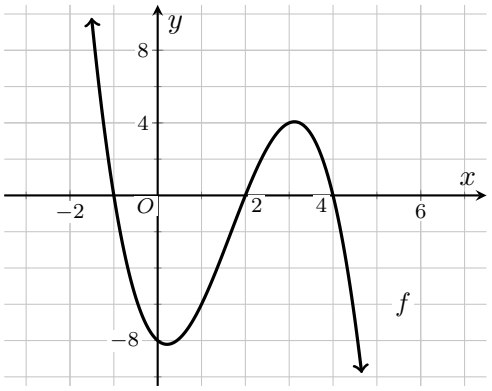
Word bank: decreasing; increasing; negative; positive.

f is _____ on the intervals $(-\infty, a)$ and (b, ∞).

f is _____ on the intervals $(-1, 2)$ and $(4, \infty)$.

f is _____ on the interval (a, b).

f is _____ on the intervals $(-\infty, -1)$ and $(2, 4)$.



Scoring: Question 2 and Question 3 are 10 points each. Question 4 is a bonus of up to 5 points. Total out of 20.

3-1 Lesson Quiz, short form: answer key

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2. A, E. $f(x) = x^3 - 7x^2 + 10x$ is already in standard form: degree 3 (odd), leading coefficient 1 (positive), so the ends point opposite ways and the right end goes up: as $x \to \infty$, $y \to \infty$; as $x \to -\infty$, $y \to -\infty$. Savvas quiz item 2, DOK 1, Skills Review and Practice A73 (taught by Example 2). **10 points:** 5 for each correct box; take 5 off for each wrong box (B, C or D), not below 0. The degree and leading-coefficient line is the student's work and is not scored.

3. B, D, G ($x = 0$, $x = -1.5$, $x = 1.5$). The graph changes from increasing to decreasing at the relative maximum $(0, 4)$ and from decreasing to increasing at the relative minimums near $(-1.5, -2)$ and $(1.5, -2)$. Choices E and F (± 1) are x-intercepts, not turning points; C (4) is the y-value of the maximum. Savvas quiz item 3, DOK 2, A73 (taught by Example 3). **10 points:** 3 for each correct box plus 1 for no wrong box; take 3 off for each wrong box, not below 0.

4. decreasing; negative; increasing; positive. Increasing/decreasing: read left to right between the turning points at $x = a = \frac{5-\sqrt{19}}{3} \approx 0.21$ and $x = b = \frac{5+\sqrt{19}}{3} \approx 3.12$. Positive/negative: above or below the x-axis between the zeros -1, 2, 4. Savvas quiz item 4, DOK 2, A73 (Example 3 for increasing/decreasing; the packet's Practice for positive/negative). **Bonus 5:** 1 point per sentence, plus 1 for all four. *Teacher note:* the Savvas quiz prints the intervals as $\left(-\infty, \frac{1}{3}\right)$, $(3, \infty)$ and $\left(\frac{1}{3}, 3\right)$, which are not exact ($f$ is already increasing at $x = 0.25$); this form names the turning points a and b instead. The answers are the same.

Graphs. Question 3 is redrawn from the TE description (W-shaped quartic, relative maximum $(0, 4)$, relative minimums at $(\pm 1.5, -2)$, x-intercepts near ± 1 and ± 2; the curve drawn is $y = \frac{32}{27}x^4 - \frac{16}{3}x^2 + 4$, whose intercepts are ± 0.98 and ± 1.88). Question 4 is the given function on the TE window; its turning points are at $x \approx 0.21$ (minimum about -8.21) and $x \approx 3.12$ (maximum about 4.06); the TE's figure note says "near $(\frac{1}{3}, -8.3)$" and "at $(3, 4)$".

Items left off this form (available for a retake or Form B): Savvas item 1, "Which statement is true of $f(x) = x^2 + 6x^3 - 4 + 2x^5$?" (answer C, the graph has exactly one x-intercept; DOK 2, A73; it needs a graph or table of a quintic, and choice D, "increasing for all real numbers," is false only on a tiny interval, about $-0.11 < x < 0$, that a hand sketch cannot show); Savvas item 5, the water-tank y-intercept, $y = 1.15x^3 - 0.1x^2 + 2$ (answer: 2; volume of water; turned on; DOK 2, A73; taught by Example 5, which this lesson cuts).

IXL replacement: SmartScore 70 or above on XMK and XJU replaces this quiz with 20/20.